

Comparison Study for RFOC and SFOC in PMSMs

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Abstract—This paper presents a comparison study between the Rotor Field-Oriented Control (RFOC) and Stator Field-Oriented Control (SFOC) for Permanent-Magnet Synchronous Motors (PMSMs). Both methods fall within the well-known Field-oriented control (FOC) category. To evaluate both methods for PMSM control, different methodologies should be provided, beginning with the control design, the Maximum Torque per Ampere (MTPA) control strategy, and the flux weakening approach. Finally, comparison between both methods based on simulation results on a 2.2kW motor model is presented.

Index Terms—Field-oriented control (FOC), Stator Flux-Oriented Control (SFOC), Rotor Flux-Oriented Control (RFOC), and Permanent Magnet Synchronous Motor (PMSM).

I. INTRODUCTION

Permanent Magnet Synchronous Motors (PMSMs) are widely employed across all industrial applications, with a steadily increasing market share over the last decades, especially in the automotive field [1]. Because the magnetic field is created by the magnets, the absence of rotor currents minimizes motor losses and temperature, resulting in more torque and power density, as well as a higher power factor, and thus better efficiency.

PMSM control methods can be divided into scalar and vector control (see Fig. 1). In scalar control, only the amplitude and frequency (angular speed) of voltage, current, and flux linkage space vectors are regulated based on steady-state equations. As a result, the scalar control has no effect on the position of the space vector during transients. In contrast, in vector control, not only the magnitude and frequency are adjusted, but also the instantaneous values of voltage, current, and flux space vectors [2], which allows transient operation. By controlling the space vectors in both steady-state and transient conditions, vector control has become the common control strategy used for the high-performance control of PMSMs [2].

Vector control is a generic control philosophy that may be applied differently. Direct Torque Control (DTC) [3] is a vector control approach that works in the space vector coordinates. Since it was introduced in 1985 [4], there has been continuous research and improvement in this method. In DTC the current

vector control is replaced with two hysteresis controllers (bang-bang control), one for the stator flux magnitude and the other for the electromagnetic torque. Furthermore, the absence of Pulse Width Modulation (PWM) in DTC results in a variable switching frequency, which is one of the disadvantages of this method. Other types of vector control methods can be used to achieve the decoupling of PMSM equations. This gave rise to nonlinear control approaches [5], such as feedback linearization control (FLC) [6] or input-output decoupling [7].

Field-Oriented Control (FOC) introduced by Blaschke [8] is the most well-known technique that has been put in place for high-performance AC motor control. Field orientation aims to reestablish the decoupling of the flux and torque-generating components of the stator current vector, as in the case of a Direct Current (DC) motor. This control approach is based on closed-loop control of the decoupled current components. To achieve this decoupling, the machine's equations are transformed into a coordinate system that rotates in synchronism with the rotor flux vector. The FOC was presented first in the rotor coordinates (RFOC), while the stator flux orientation (SFOC) scheme was developed later on to avoid the parameter sensitivity of RFOC in the case of rotor flux estimation for induction motors [9]. Recently, the SFOC was suggested to control PMSMs mainly at high-speed applications in the flux weakening region [10]. In the literature, nonlinear controllers are equally used to improve the performances of both SFOC and RFOC approaches [11], [12], [13].

This paper presents a formal comparative study of the PMSM FOC approach, particularly RFOC and SFOC based on linear PI controllers. Indeed, RFOC based on linear PI controllers is the most widely implemented vector controller in industrial applications. Thus, to keep the comparison as fair as possible, we compare it with SFOC based on PI controllers. Section II presents the conventional model of the PMSM in the rotor- and stator-synchronous reference frames. The control design study for each method was performed in Section III. Furthermore, the Maximum Torque Per Ampere (MTPA) control strategy is studied in Section IV, while the flux weakening analysis is investigated for both methods in

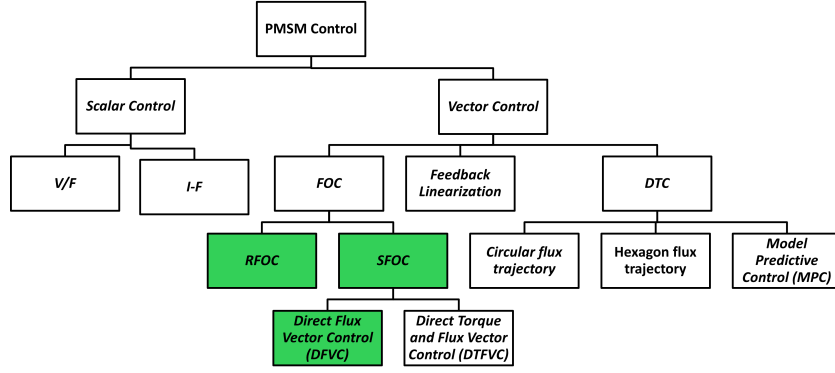


Fig. 1. Classification of PMSM control methods

Section V. Finally, the simulation results, presented in Section VI, show the performance of each method when applied to a 2.2 kW PMSM model.

II. PMSM MODEL

In order to simplify the PMSM model, different assumptions are made:

- The magnetic circuit is linear (no saturation).
- Hysteresis and Foucault's currents are neglected.

A. Motor model in rotor flux coordinates

The rotor flux-oriented (RFO) frame of PMSMs is based on the rotor flux orientation [14], which is simply the direction of the magnets' north pole (see Fig. 2). From the above assumptions, the PMSM voltage equations in the RFO frame are given as [2]:

$$v_d = R_s i_d + \frac{d\psi_d}{dt} - \omega \psi_q \quad (1)$$

$$v_q = R_s i_q + \frac{d\psi_q}{dt} + \omega \psi_d \quad (2)$$

where v , i , and ψ denote the voltage, current, and flux, respectively. The stator resistance is defined by R_s , while ω represents the electrical angular frequency. The PMSM flux-current equation is written as:

$$\psi_d = L_d i_d + \psi_m \quad ; \quad \psi_q = L_q i_q \quad (3)$$

such that L_d and L_q are the dq -axis inductances, respectively. ψ_m is the permanent magnet flux. The PMSM electromagnetic torque T_e equation is:

$$T_e = \frac{3}{2} n_p (\psi_d i_q - \psi_q i_d) = \frac{3}{2} n_p (\psi_m + (L_d - L_q) i_d) i_q \quad (4)$$

where n_p is the number of pole pairs of the machine. The above equation can be simplified for Surface PMSM (SPMSM), where the stator inductances are almost equal such that $L_d = L_q = L_s$.

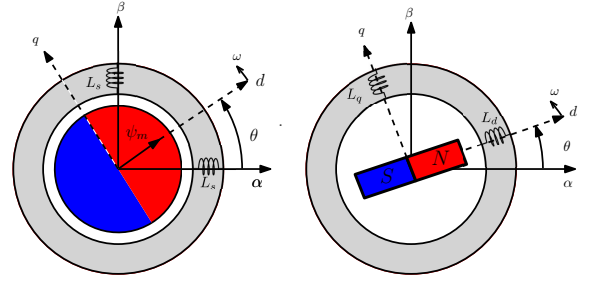


Fig. 2. SPMSM (left) and IPMSM (right) rotor flux reference frame

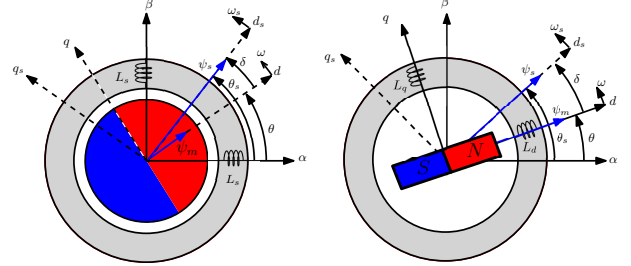


Fig. 3. Stator flux reference frame

B. Motor model in stator flux coordinates

The stator flux-oriented (SFO) frame rotates at a synchronous speed ω , such that the direct axis follows the stator magnetic flux vector $\bar{\psi}_s$ rather than the rotor magnetic flux $\bar{\psi}_m$ (see Fig. 3) [15]. The mechanical position of the rotor corresponds with the rotor flux vector. The angle δ between the stator and rotor flux vectors is defined as the load/torque angle. In the SFO frame, the stator flux is the d_s -axis flux:

$$\psi_{d_s} = \psi_s \quad ; \quad \psi_{q_s} = 0 \quad (5)$$

where ψ_s is the amplitude of stator flux vector $\bar{\psi}_s$. Writing the voltage equations in the SFO reference frame leads to:

$$v_{d_s} = R_s i_{d_s} + \frac{d\psi_s}{dt} \quad (6)$$

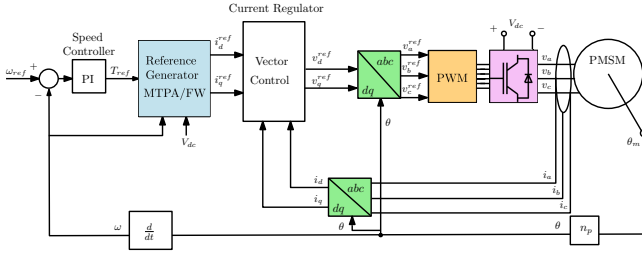


Fig. 4. RFOC complete control structure

$$v_{q_s} = R_s i_{q_s} + (\omega + \frac{d\delta}{dt}) \psi_s \quad (7)$$

From (4) and (5) the torque equation in the SFO frame is then obtained as:

$$T_e = \frac{3}{2} n_p \psi_s i_{q_s} \quad (8)$$

III. RFOC AND SFOC CONTROL DESIGN

In this section, the controller design for both RFOC and SFOC is developed.

A. RFOC

The RFOC of PMSMs is done in the RFO frame [16]. From (1) and (2), we obtain the dq currents state space equations:

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} i_d \\ i_q \end{bmatrix} &= \begin{bmatrix} -\frac{R_s}{L_d} & \omega \frac{L_q}{L_d} \\ -\omega \frac{L_d}{L_q} & -\frac{R_s}{L_q} \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} \\ &+ \begin{bmatrix} \frac{1}{L_d} & 0 \\ 0 & \frac{1}{L_q} \end{bmatrix} \begin{bmatrix} v_d \\ v_q - \omega \psi_m \end{bmatrix} \end{aligned} \quad (9)$$

For controlling the currents in RFOC, a Proportional Integral (PI) current control can be implemented. There are mainly two types of PI controllers used in the literature: synchronous PI current regulator with cross-coupling decoupling and complex vector synchronous PI with active damping [17]. This study will focus on the first approach due to its similarities with the SFOC controller structure. The complete scheme for RFOC is shown in Figs. 4 and 5, such that the PI coefficients are defined as follows:

$$K_{pd} = \omega_{bc} L_d \quad ; \quad K_{pq} = \omega_{bc} L_q, \quad (10)$$

$$K_{id} = K_{iq} = \omega_{bc} R_s, \quad (11)$$

where ω_{bc} is the chosen current controller bandwidth.

B. SFOC: Direct Flux Vector Control (DFVC)

DFVC is a control technique that utilizes the SFO frame [12], [18]. In DFVC rather than controlling the dq currents, we control the torque variables which are the stator flux magnitude ψ_s and the torque-producing current component i_{q_s} . For the simplicity of control, it is important to obtain the state equation of i_{q_s} . The vector coordinates transformation from the rotor to stator coordinates is defined as follows:

$$\begin{bmatrix} i_{d_s} \\ i_{q_s} \end{bmatrix} = \begin{bmatrix} \cos \delta & \sin \delta \\ -\sin \delta & \cos \delta \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} \quad (12)$$

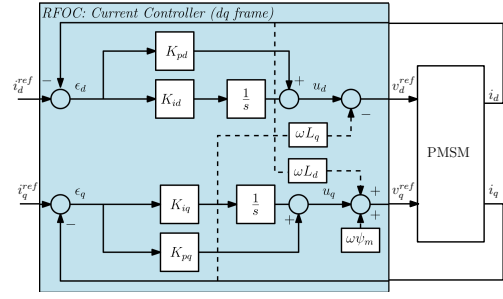


Fig. 5. PMSM PI current regulator in RFOC, shown in a rotor synchronous reference frame. Dashed lines correspond to feedforward terms

Using (12) the q_s -axis equation can be written as [10]:

$$L_d \frac{di_{q_s}}{dt} = -k(\delta)(v_{d_s} - R_s i_{d_s}) + b(\delta, \psi_s)(v_{q_s} - R_s i_{q_s} - \omega \psi_s) \quad (13)$$

The two factors $k(\delta)$ and $b(\delta, \psi_s)$ are defined below:

$$k(\delta) = (1 - \frac{L_d}{L_q}) \frac{\sin 2\delta}{2}; \quad (14)$$

$$b(\delta, \psi_s) = -(1 - \frac{L_d}{L_q}) \cos 2\delta + \frac{\psi_m \cos \delta}{\psi_s} \quad (15)$$

Note that for SPMSM, $k(\delta) = 0$ and $b(\delta, \psi_s) = \frac{\psi_m \cos \delta}{\psi_s}$. The obtained nonlinear model of both i_{q_s} and ψ_s is defined as:

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} \psi_s \\ i_{q_s} \end{bmatrix} &= \begin{bmatrix} -\frac{R_s}{L_d} & 0 \\ \frac{k(\delta)R_s}{L_d} & -\frac{b(\delta, \psi_s)R_s}{L_d} \end{bmatrix} \begin{bmatrix} i_{d_s} \\ i_{q_s} \end{bmatrix} \\ &+ \begin{bmatrix} \frac{1}{L_d} & 0 \\ -\frac{k(\delta)}{L_d} & \frac{b(\delta, \psi_s)}{L_d} \end{bmatrix} \begin{bmatrix} v_{d_s} \\ v_{q_s} \end{bmatrix} + \begin{bmatrix} 0 \\ -\frac{b(\delta, \psi_s)}{L_d} \omega \psi_s \end{bmatrix} \end{aligned} \quad (16)$$

In contrast to RFOC, DFVC has only one current control loop. However, the current loop dynamics are nonlinear and coupled with d_s -axis voltage, thus derating the control performance. To control ψ_s and i_{q_s} , both linear and nonlinear controllers [13] were proposed in the literature. In this study, we choose to compare RFOC with the SFOC scheme that uses two PI controllers. This renders the comparison fair since both approaches have the same structure. Of course, from (16), we can see that the tuning of PI controllers will depend on the nonlinear functions $b(\delta, \psi_s)$ and $k(\delta)$. Different tuning methods, such as designing the controllers by considering the best case (b_{max}), or taking into account the variations of $b(\delta, \psi_s)$ and $k(\delta)$ by adaptive PI controllers, can be used. In this paper, we consider that the last tuning approach is far from what is used in the traditional RFOC scheme. Moreover, adaptive controller implementation may introduce some undesirable computational complexity in industrial applications. Thus, here, the PI controllers in SFOC are tuned according to the first tuning strategy in which the tuning of PI coefficients depends on the two factors $k(\delta)$ and $b(\delta, \psi_s)$. The complete structure of DFVC is shown in Fig. 6.

1) *Flux ψ_s Regulator*: The flux loop can be controlled to respond very quickly using v_{d_s} according to (6) with no coupling effect from v_{q_s} . Aside from the discretization constraint

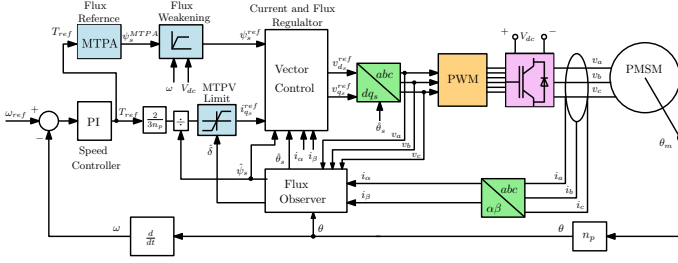


Fig. 6. SFOC complete control structure

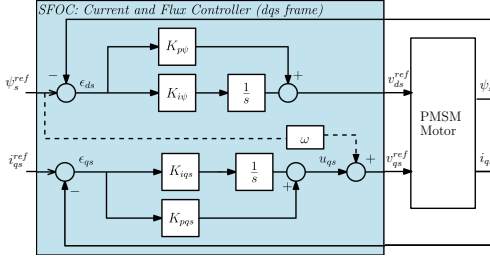


Fig. 7. PMSM PI current and flux regulator in SFOC, shown in a stator synchronous reference frame. Dashed lines correspond to feedforward terms

caused by PWM, the flux bandwidth ω_{bf} is restricted only by the dynamics of the flux observer.

2) *Current i_{qs} Regulator*: To control i_{qs} a fast PI regulator is needed to cancel the cross-coupling effect from the flux loop (16), whereas the back-EMF is compensated in a feed-forward manner. The dynamic response of i_{qs} is influenced by $b(\psi_s, \delta)$ (see (15)), which fluctuates with the motor's operating point. The vector control for both i_{qs} and ψ_s is shown in Fig. 7, with PI coefficients defined as follows:

$$K_{p\psi} = \omega_{bf} \quad ; \quad K_{p i_{qs}} = \omega_{bc} \frac{L_d}{b_{max}}, \quad (17)$$

$$K_{i\psi} = \omega_{bf} R_s \quad ; \quad K_{i i_{qs}} = \omega_{bc} R_s, \quad (18)$$

where b_{max} is the maximum value of $b(\psi_s, \delta)$.

3) *Maximum Torque Per Volt (MTPV) Limit*: The control of i_{qs} is not stable in the case $b(\psi_s, \delta) < 0$, due to positive feedback [10]. To prevent this instability, this constraint is respected by the limitation of i_{qs}^{ref} by a saturation limit.

4) *Flux Observer*: The adopted observer in our study is based on both the current and voltage models [19]. From the flux observer, we can obtain both the stator flux angle $\hat{\theta}_s$ and magnitude $\hat{\psi}_s$.

IV. MTPA

The MTPA control strategy is an important factor in the torque generation of the Interior PMSM (IPMSMs) [20].

A. Maximum Torque per Amper (MTPA) in RFOC

To obtain the maximum efficiency in RFOC from the motor the current references must follow the MTPA equations when possible, such that i_{dq}^{ref} should be computed as follows [20]:

$$i_{d_{MTPA}} = \frac{\psi_m - \sqrt{\psi_m^2 + 8L_d^2 I_s^2}}{-4L_\Delta} < 0 \quad (19)$$

where $L_\Delta = L_d - L_q < 0$

$$i_{q_{MTPA}} = \text{sign}(T_{ref}) \sqrt{I_s^2 - i_{d_{MTPA}}^2} \quad (20)$$

such that the current vector magnitude I_s is defined as follows:

$$I_s = \frac{2|T_{ref}|}{3n_p \psi_m} \quad (21)$$

Note that for SPMSM in RFOC, the MTPA control strategy is obtained by setting $i_{d_{MTPA}} = 0$ and $i_{q_{MTPA}} = \text{sign}(T_{ref}) I_s$.

B. MTPA in SFOC

The MTPA trajectory in SFOC is defined as the control points for achieving the maximum torque at any given flux [10]. In contrast to RFOC, SFOC controls directly the flux to produce the maximum torque. The flux reference must be set according to the MTPA curve to find the optimal flux values for a given torque.

V. FLUX WEAKENING (FW)

A motor can be accelerated by the maximum torque until the terminal voltage reaches its maximum value V_{max} , at this point, we obtain the maximum speed, defined as ω_{base} [21].

A. FW in RFOC

In the case of RFOC, the voltage and current limited maximum torque control (VCLMT) [22] is the most employed method in the feedforward FW technique. In this method, the i_d current reference is computed as follows [23]:

$$i_{d_{fw}} = \frac{-\psi_m L_d + \Delta}{L_d^2 - L_q^2} < 0 \quad (22)$$

where Δ is defined as:

$$\Delta = \sqrt{(\psi_m L_d)^2 - (L_d^2 - L_q^2)(\psi_m^2 + L_q I_s^2 - V_{max}^2/\omega^2)} \quad (23)$$

For IPMSM, i_d is always negative as long as $L_d < L_q$. Where i_{dq}^{ref} depends on the motor speed, which can be working either in the MTPA or FW region. If $\omega < \omega_{base}$:

$$i_d^{ref} = i_{d_{MTPA}} \quad ; \quad i_q^{ref} = i_{q_{MTPA}} \quad (24)$$

Above the base speed, the i_d and i_q for MTPA control cannot fulfill the voltage constraint. If $\omega > \omega_{base}$ then:

$$i_d^{ref} = i_{d_{fw}} \quad ; \quad i_q^{ref} = \text{sign}(T_{ref}) \sqrt{I_s^2 - i_{d_{fw}}^2} \quad (25)$$

B. FW in SFOC

The FW in the SFOC is simpler than in the RFOC and may be controlled directly via the flux ψ_s rather than i_d [24]. From (7) the flux reference is limited according to:

$$\psi_s^{ref} \leq \frac{V_{max}}{|\omega|} \quad (26)$$

And by adding the resistance drop we obtain:

$$\psi_s^{ref} \leq \frac{\sqrt{V_{max}^2 - (R_s i_{ds})^2} - R_s i_{qs}}{|\omega|} \quad (27)$$

VI. SIMULATIONS AND PERFORMANCE COMPARISON

Both RFOC and SFOC were tested in simulation, using Matlab/Simulink, on the 2.2kW-motor model whose parameters are presented in Table I. To set up the simulation environment the current controller bandwidth is fixed to $\omega_{bc} = 2\pi 100$ rad/s. While the flux loop is chosen two times faster than the current controller, so that $\omega_{bf} = 2\pi 200$ rad/s. The PI controllers of both RFOC and DFVC are tuned according to the motor parameters which are assumed to be known.

A. Current control loops performance

To evaluate the performance of current controllers for both methods a step response (0.5 p.u) test was done at standstill on both i_q and i_{qs} (Fig 9). It is obvious that RFOC has no overshoot and a faster settling time ($t_s = 5$ ms) compared to SFOC i_{qs} current control ($t_s = 10$ ms). This is due to the fact of the non-linearity between v_{qs} and i_{qs} . To test the sensitivity of the current controllers, the dq -axis inductances are altered. In the case of the RFOC, the q -axis current loop is tested by changing L_q by $\pm 30\%$, which affected the settling time with a small increase as well as a minor overshoot (see Fig. 10). As for SFOC, since i_{qs} control depends only the d -axis inductance (13), by decreasing L_d by 30% the settling time t_s increased from 10ms to 16ms (Fig. 11).

TABLE I
2.2kW PMSM MOTOR NAMEPLATE

PMSM Motor 2.2kW	
Rated Voltage	400 [V]
Rated Speed	1500 [rpm]
Rated Current	4.3 [A]
Rated Frequency	75 [Hz]
R_s	3.6 [Ω]
L_d	36 [mH]
L_q	51 [mH]
ψ_m	0.545 [Vs]
J	0.015 [$kg.m^2$]

B. Torque control performance

To compare both methods, different operating points (A, B, C, D, E) are chosen in the speed-torque plane (Fig. 8). Table II shows the results for torque control performance criteria: rising time (t_r), settling time (t_s), and percentage overshoot (PO). It is shown that although both methods have been tuned to have the same bandwidth, RFOC has a better dynamic response than SFOC in which the rising and settling times for the torque control are faster in all the tested operating points. This difference becomes more obvious as the torque reference increases, leading to different operating points with larger torque angles. Fig. 15 shows the speed control in the FW region (point E), where it is seen that RFOC is slightly faster than SFOC. In Fig. 16 the ψ_s and i_d control in FW are shown. It is seen that both methods achieve almost the same tracking error in transient periods for the i_d current and ψ_s control. The MTPA control strategy was tested for both methods (Fig. 17), where SFOC shows a slightly better performance concerning

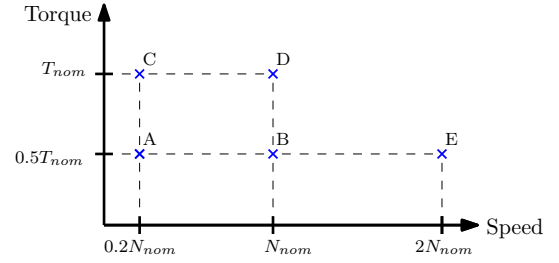


Fig. 8. Speed-torque profile operating test points

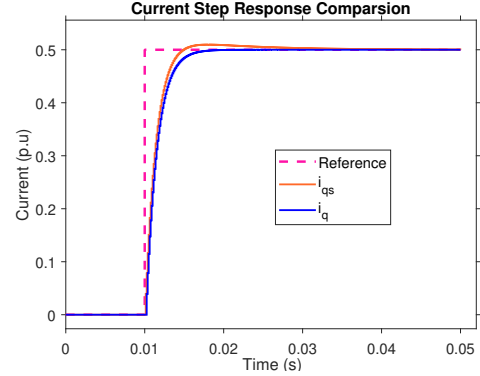


Fig. 9. Step current response comparison between RFOC and SFOC

I_s as the load torque increases, thus achieving better efficiency.

TABLE II
TORQUE CONTROL PERFORMANCE ON DIFFERENT OPERATING POINTS

Pts	RFOC			SFOC		
	t_r (ms)	t_s (ms)	PO (%)	t_r (ms)	t_s (ms)	PO (%)
A	3.15	5.1	0	3.2	9	1.5
B	3.25	5.2	0	3.4	11	2.5
C	3.15	5.2	0.5	3.3	10	2.5
D	3.2	5.25	0.5	3.7	10	2.7
E	3.3	7	3	4.1	12	5

VII. CONCLUSION

This paper presents a comparative study of RFOC and SFOC for PMSM. It is demonstrated that RFOC delivers a faster transient speed response than SFOC while keeping the control system simpler. Although the SFOC scheme is disadvantaged by parameter sensitivity and a more complex tuning to the current controller, it is seen that SFOC is advantaged by a less parameter-dependent FW control strategy. When compared with RFOC, the SFOC approach for PMSM might be viewed as a novel strategy that can be improved in terms of performance by using more advanced controllers. Finally, future studies should be focused on validating the simulations on a real PMSM hardware setup.

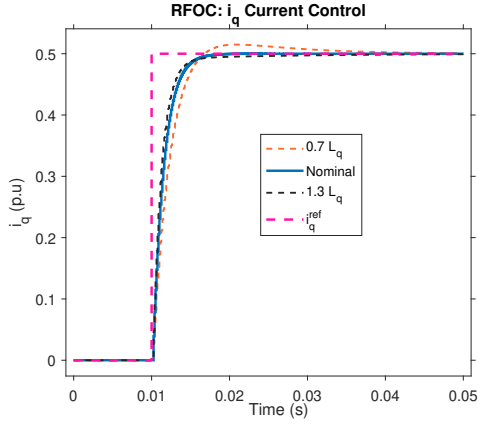


Fig. 10. L_q error impact on i_q control response (RFOC)

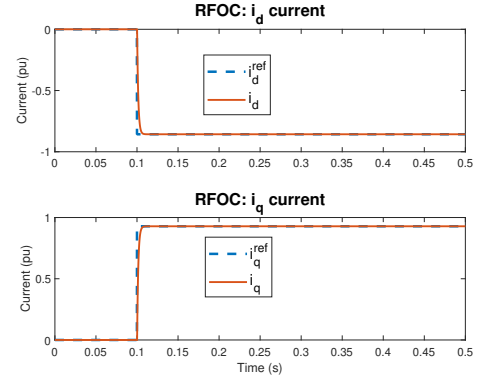


Fig. 13. RFOC current control under nominal torque reference at nominal speed (point D)

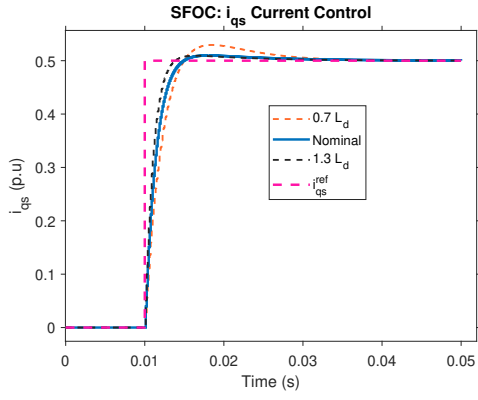


Fig. 11. L_d error impact on i_{qs} control response (SFOC)

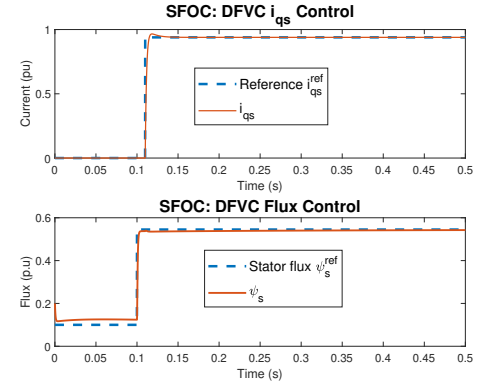


Fig. 14. SFOC flux and current control under nominal torque reference at nominal speed (point D)

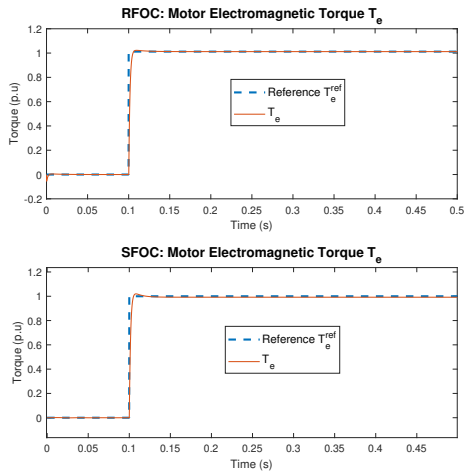


Fig. 12. Torque response for RFOC and SFOC at nominal speed (point D)

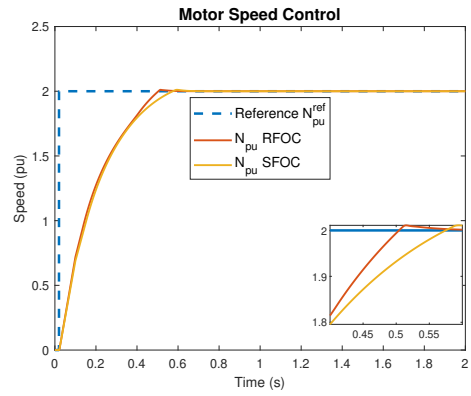


Fig. 15. Speed response in FW region (point E)

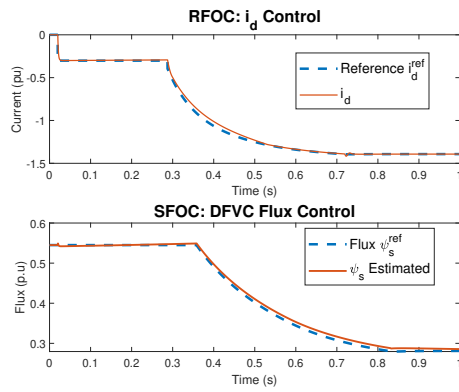


Fig. 16. ψ_s and i_d control in FW region (point E)

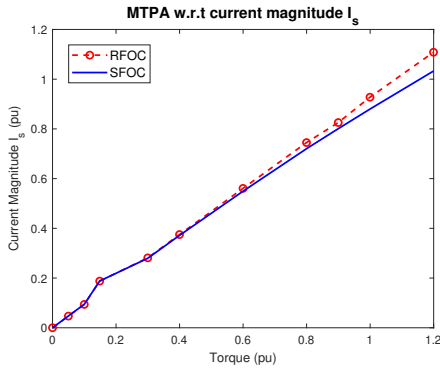


Fig. 17. MTPA in terms of current magnitude I_s for both RFOC and SFOC

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